Kakali Bandyopadhyay

PE Value-Creation and Returns Model | Personal Prototype on Synthetic Data

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The question

Does a five-year hold of a synthetic mid-market B2B services company clear an illustrative 20% IRR hurdle, and how much of the return depends on value-creation initiatives rather than core growth?

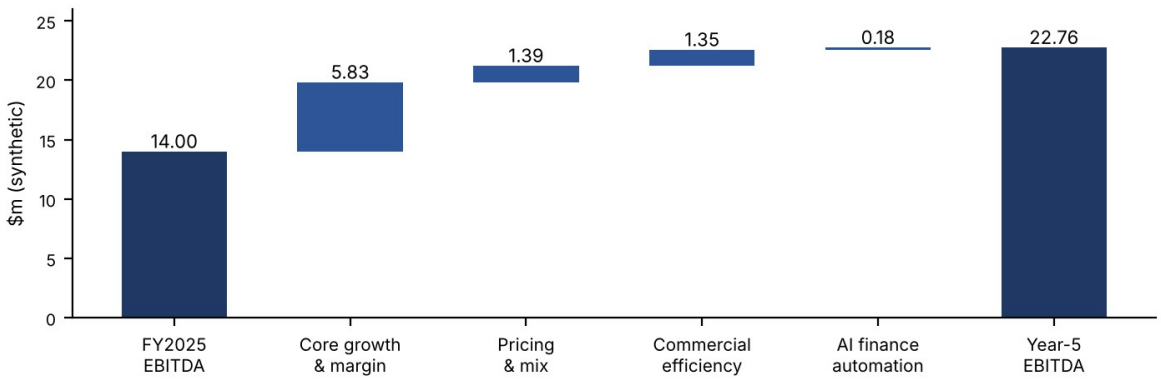
What I built

- **Live-formula Excel model:** Base / Downside / Upside scenarios that change growth, margin, initiative timing, realisation and costs, not just the exit multiple; entry structure, debt schedule with amortisation and cash sweep, tax, capex and working capital; MOIC and IRR from dated cash flows; two sensitivity grids.
- **Three toggleable value levers:** pricing and mix, commercial efficiency, AI finance automation, each with owner, start year, ramp, implementation and run cost, realisation rate and dependency.
- **Execution and controls:** quarterly value-creation plan tracker with visible RAG thresholds and an expand / fix / pause gate for the AI lever; 12 integrity checks; returns recomputed independently in Python across all scenarios; 17 automated tests.

Returns ($m, synthetic)

Scenario	Entry equity	Exit EBITDA	Exit multiple	Exit equity	MOIC	IRR	Checks
Base	**86.8**	**22.76**	**10.0x**	**210.1**	**2.42x**	**19.3%**	**12/12 PASS**
Downside	86.8	17.24	8.5x	116.8	1.35x	6.1%	12/12 PASS
Upside	86.8	25.67	11.0x	271.7	3.13x	25.6%	12/12 PASS

Base EBITDA bridge: FY2025 to exit year ($m, synthetic)



What the model shows and the decision it supports

- **Core drives the case:** core growth and margin supply about two-thirds of the $8.76m EBITDA increase; the three initiatives supply about one-third ($2.92m), of which AI is only $0.18m.
- **Just short of the hurdle:** Base IRR of 19.3% misses an illustrative 20% hurdle; closing the gap needs roughly $0.6m more exit EBITDA or about 0.26x on the exit multiple. Decision: proceed only if diligence can evidence the core growth plan.
- **AI benefit is counted only when realised:** 16,200 hours released in year 5 are worth $0.59m before realisation, but only $0.38m of avoided cost reaches EBITDA. Redeployed hours are not treated as earnings.
- **Downside test:** the AI lever turns net negative (run cost exceeds realised savings); switching it off lifts Downside MOIC from 1.35x to 1.37x, which is the case for the Pause gate.

All companies, figures, targets and tracker actuals are synthetic and illustrative. Personal prototype; not a real deal, investment advice or a claim of realised value.